

Divide and Conquer

The depth of the tree $\rightarrow$ number of comparisons in the longest path from root to leaf $\rightarrow$ worst-case time complexity!

$$n \text{ elements} \rightarrow \{1, 2, \dots, n\}$$

Number of leaf nodes $\rightarrow n!$ (# of permutations of n elements)

Number of leaf nodes in a (binary) tree of depth $d \rightarrow 2^d$

So:

$$2^d \geq n! \rightarrow d \geq \log_2(n!) \quad \int_{(i+1)(n-i), i=0 \dots \frac{n}{2}-1}$$

$$n! = 1 \cdot 2 \cdot 3 \cdot \dots \cdot n = (1 \cdot n)(2 \cdot [n-1])(3 \cdot [n-2]) \dots \left(\frac{n}{2} \cdot \left[\frac{n}{2} + 1\right]\right)$$

Consider these if n is odd

$$\geq \left(\frac{n}{2}\right) \left(\frac{n}{2}\right) \dots \left(\frac{n}{2}\right) = \left(\frac{n}{2}\right)^{\frac{n}{2}}, n \geq 1$$

$$d \geq \log_2(n!) \geq \frac{n}{2} \log_2 \left(\frac{n}{2}\right) = An \log_2(n) - Bn$$

Complexity $\rightarrow$

$$\Omega(n \log n)$$

Finding Median*

$$A = [A_1, A_2, A_3, \dots, A_n]$$



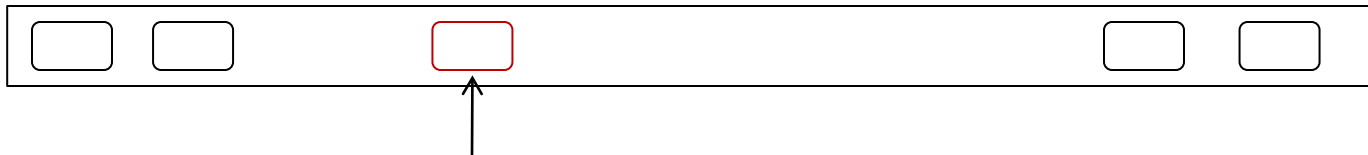
Output k^{th} smallest entry in A . Median $k = \frac{n}{2}$

Sort first to get $A_s = [A_{(1)}, A_{(2)}, A_{(3)}, \dots, A_{(n)}]; A_{(i)} \leq A_{(j)}, i < j$
 $\downarrow$
 $A_{(\frac{n}{2})}$

Using only comparisons (Mergesort) $O(n \log(n))$

Linear time??

Brief on quick select (**randomized** algo):



Randomly choose a pivot, compare, all smaller to left, all larger to right; then choose one side (based on pivot index) and repeat the same.

Ideally... (with the pivot is always the median)

$$T(n) = T\left(\frac{n}{2}\right) + Cn \longrightarrow \# \text{ comparisons with the pivot}$$

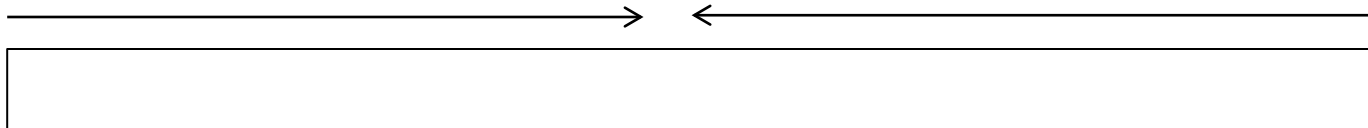
$$a = 1, b = 2, d = 1 \implies O(n)$$

and bad luck... (with the pivot is always the smallest / largest)

$$n + (n - 1) + (n - 2) + \cdots + \frac{n}{2} = \Theta(n^2) \quad \text{For median}$$

Worse to better pivot? ...

... better to worse pivot?



With uniform random selection.. even if we have pivots at 10th percentile....

$$T(n) = T\left(\frac{9n}{10}\right) + Cn$$

a random variable
Highly likely?!

$$a = 1, b = 10/9, d = 1 \implies O(n)?$$

Can we design a deterministic algo, which ensures that the pivot will be somewhere in the middle?

A simplifying WLOG **reduction**...

Consider A doesn't have duplicate entries!

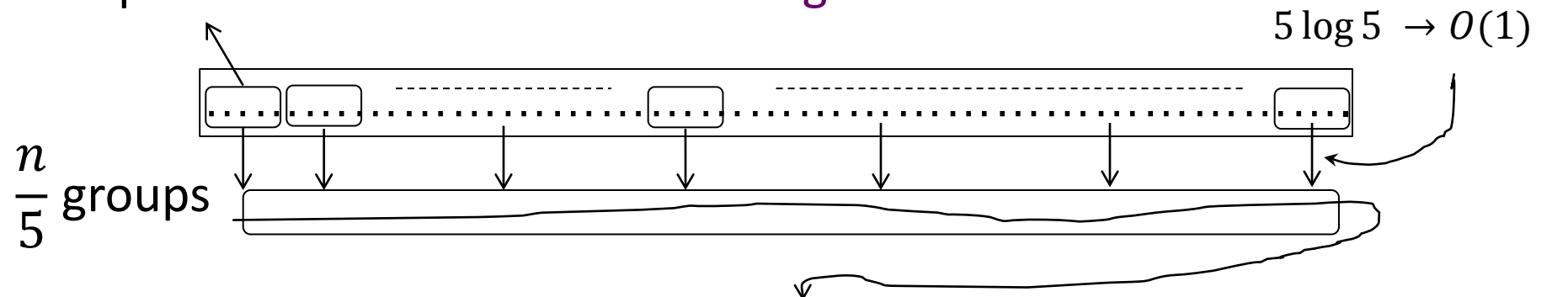
$$A = [A_1, A_2, A_3, \dots, A_n]$$

We ensure by $B = [(A_1, 1), (A_2, 2), (A_3, 3), \dots, (A_n, n)]$

Comparison only between first elements, and then the second only when required.

Groups of 5

The select algorithm



'Median' of medians as pivot $\rightarrow$ Using the select algo!

Select ($A = [A_1, A_2, A_3, \dots, A_n], k$)

1. Break A into groups having 5 elements each.
2. $B \leftarrow$ Array of medians from all the groups.
3. $P \leftarrow \text{Select} \left(B, \frac{n}{10} \right) \text{ or } \left(B, \frac{\frac{n}{5} + 1}{2} \right)$
4. $L \leftarrow$ set of elements in $A < P$, $R \leftarrow$ set of elements in $A > P$
- 5(a). If $k = |L| + 1$: return P
- 5(b). elseif $k < |L| + 1$: return $\text{Select}(L, k)$
- 5(c). else $k > |L| + 1$: return $\text{Select}(R, k - |L| - 1)$

Runtime...

$$T(n) = \overset{1}{O(n)} + \overset{2}{\frac{n}{5} O(1)} + \overset{3}{T\left(\frac{n}{5}\right)} + \overset{4}{O(n)} + \overset{5}{T\left(\frac{7n}{10}\right)}$$

Maximum
size of L / R

$$T(n) = T\left(\frac{n}{5}\right) + T\left(\frac{7n}{10}\right) + O(n)$$

P is $\gtrless$ than at least

$$\frac{n}{5} \cdot \frac{1}{2} \cdot 3 = \frac{3n}{10}$$

Lets prove by induction, guess and check ... that this accurate algo is linear in n . We will also see the significance of 5!



Any odd integer greater than or equal to 5.

Guess:

$$T(n) \leq Dn$$

Is this true?

Start with $n = 1: O(1)$, will be true for some finite constant D



$n \leq 5$ will be true for D as some constant

$n \uparrow$ Induction:

If



$$D \frac{n}{5}$$

$$+ D \frac{7n}{10}$$

$$+ Cn \leq Dn$$

Is this true?

Consider

$$\frac{D}{5} + \frac{7D}{10} + C \leq D \rightarrow \frac{9D}{10} + C \leq D \quad \text{is true because we can take}$$

$$C \leq \frac{D}{10} !$$

Check: 3 per group does not work!